

mdspan.at()

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§ 1 Introduction

This paper proposes element access with bounds checking to `std::mdspan` via `at()` member functions.

§ 2 Revision history

§ 2.1 R3

- Swap paragraph 8 and 9 and use I instead of indices

§ 2.2 R2

- Remove feature test macro bump requested by LEWG
- Adapt wording to follow same pattern like `operator[]` and rebase on top of latest draft

§ 2.3 R1

Update wording to align with [P2821R5]:

- Bump `__cpp_lib_mdspan` instead of adding `__cpp_lib_mdspan_at`
- Mark `at()` as freestanding deleted and adjust freestanding comments of header and class

§ 2.4 R0

Initial version

§ 3 Motivation

§ 3.1 Safety

The new `at()` member functions provide memory-safe element access to `std::mdspan`, and thus have defined behavior. Out-of-bound access can be caught by catching the `std::out_of_range` exception.

§ 3.2 Consistency

In [P2821R5], element access with bounds checking via `at()` has been added to `std::span`. One of the main motivations for this change was consistency with other containers that have element access with bounds checking via `at()`. Similarly, such element access should be added to `std::mdspan`.

§ 4 Impact On the Standard

The impact of this proposal on the standard is low. The proposed function signatures for the `at()` member functions are identical to the function signatures for the subscript operators as proposed in [P0009R18].

One consideration is that the `at()` method has previously not been used with multi-dimensional arguments. However, this was also true for the subscript operator before the possibility was introduced in [P2128R6].

§ 5 Wording

The wording is relative to [N5001].

In 23.7.3.2 ([mdspan.syn]), change the following lines:

```
- // all freestanding
+ // mostly freestanding
namespace std {

// 23.7.3.6, class template mdspan
template<class ElementType, class Extents, class LayoutPolicy =
    layout_right,
    class AccessorPolicy = default_accessor<ElementType>>
- class mdspan;
+                                     class                               mdspan;
    // partially freestanding
```

In 23.7.3.6.1 ([mdspan.mdspan.overview]), add the following immediately after the subscript operators:

```
template<class... OtherIndexTypes>
    constexpr reference at(OtherIndexTypes... indices) const;    //
    freestanding-deleted
template<class OtherIndexType>
    constexpr reference at(span<OtherIndexType, rank()> indices)
    const;    // freestanding-deleted
template<class OtherIndexType>
    constexpr reference at(const array<OtherIndexType, rank()>&
    indices) const;    // freestanding-deleted
```

In 23.7.3.6.3 ([mdspan.mdspan.members]), add the following immediately after the subscript operators:

```
template<class... OtherIndexTypes>
    constexpr reference at(OtherIndexTypes... indices) const;
```

7 Constraints:

- (7.1) — `(is_convertible_v<OtherIndexTypes, index_type> && ...)` is true,
- (7.2) — `(is_nothrow_constructible_v<index_type, OtherIndexTypes> && ...)` is true, and
- (7.3) — `sizeof...(OtherIndexTypes) == rank()` is true.

8 Let `I` be `extents_type::index_cast(std::move(indices))`.

9 *Returns:* (*this)[I...].

10 *Throws:* out_of_range if I is not a multidimensional index in extents().

```
template<class OtherIndexType>
    constexpr reference at(span<OtherIndexType, rank()> indices)
        const;
```

```
template<class OtherIndexType>
    constexpr reference at(const array<OtherIndexType, rank()>&
        indices) const;
```

11 *Constraints:*

(11.1) — is_convertible_v<const OtherIndexType&, index_type> is true,
and

(11.2) — is_nothrow_constructible_v<index_type, const
OtherIndexType&> is true.

12 *Effects:* Let P be a parameter pack such that
is_same_v<make_index_sequence<rank()>, index_sequence<P...>> is
true. Equivalent to: return at(extents_type::index-
cast(as_const(indices[P]))...);

§ 6 Feature Test Macro

LEWG decided to drop the bump of the feature test macro.

§ 6.1 Poll

Remove the feature test macro bump from P3383R1

SF	F	N	A	SA
4	4	4	2	0

§ 7 Reference Implementation

The at() member functions have been implemented in the std::mdspan reference implementation from the Kokkos project at Sandia National Laboratories [kokkos/mdspan], see [kokkos/mdspan#302].

§ 8 References

[kokkos/mdspan] Reference implementation of mdspan.

<https://github.com/kokkos/mdspan>

[kokkos/mdspan#302] Add element access via `at()` to `std::mdspan`.

<https://github.com/kokkos/mdspan/pull/302>

[N5001] Thomas Köppe. 2024-12-17. Working Draft, Programming Languages — C++.

<https://wg21.link/n5001>

[P0009R18] Christian Trott, D.S. Hollman, Damien Lebrun-Grandie, Mark Hoemmen, Daniel Sunderland, H. Carter Edwards, Bryce Adelstein Lelbach, Mauro Bianco, Ben Sander, Athanasios Iliopoulos, John Michopoulos, Nevin Liber. 2022-07-13. MDSPAN.

<https://wg21.link/p0009r18>

[P2128R6] Corentin Jabot, Isabella Muerte, Daisy Hollman, Christian Trott, Mark Hoemmen. 2021-09-14. Multidimensional subscript operator.

<https://wg21.link/p2128r6>

[P2821R5] Jarrad J. Waterloo. 2023-12-18. `span.at()`.

<https://wg21.link/p2821r5>